David Eccles School of Business
THE UNIVERSITY OF UTAH

Business Data Mining (IS 4482-001)

Fall 2026
Department of Operations & Information Systems
David Eccles School of Business, University of Utah

Course Information	Details
Instructor	Matthew Pecsok
Office Hours	See Canvas
Contact Method	Canvas
Course Type	In person
Class Meetings	Tuesdays, 6:00 p.m.-9:00 p.m.
Location	SFEBB 3106
First Class Meeting	Tuesday, August 25, 2026
Last Class Meeting	Tuesday, December 8, 2026

Course listing: University of Utah Fall 2026 Class Schedule

Fall 2026 Academic Calendar

Event	Date
Classes begin	Monday, August 24
Last day to add without a permission code / last day to wait list	Friday, August 28
Last day to add, drop (delete), elect CR/NC, or audit	Friday, September 4
Labor Day	Monday, September 7
Census deadline	Monday, September 14
Fall Break	Saturday, October 10-Sunday, October 18
Last day to withdraw from classes	Friday, October 30
Thanksgiving Break	Thursday, November 26-Sunday, November 29
Last day to reverse the CR/NC option	Friday, December 4
Classes end	Thursday, December 10

Event	Date
Reading Day	Friday, December 11
Final exam period	Monday, December 14-Friday, December 18

See the University of Utah Fall 2026 Academic Calendar for official dates and updates.

Course Description

In today's modern companies data is one of the most valuable resources. Data Mining is a process in which data is used to discover valuable patterns using techniques such as regression, clustering, classification and association rule mining. Statistical/machine learning algorithms that can discover these patterns can be used to make predictions such as whether or not a transaction is fraudulent, what the future price of a stock will be or what two items are frequently purchased together in a store. The use of these algorithms can create new valuable information for a business and can benefit many areas of an organization.

The course will cover the following data mining techniques - classification, numeric prediction, clustering and association rule mining.

Course Objectives

The objective of this course is to provide students with the data mining knowledge foundation and skills to meaningfully leverage data using machine and statistical learning methods. After taking the class, students are expected to be able to select, apply, evaluate, and understand four types of data mining tasks – classification, numeric prediction, clustering, and association rules mining for applications. To achieve these learning goals, the course will focus on the main concepts, methods, and process of data mining as well as hands-on practices using Python.

Course Materials

The primary textbook is Data Science for Business and is available within the University of Utah for free. Having free access to O'Reilly books is a great benefit to students. I highly recommend you use it to supplement your reading for other classes as well. To access the textbook use the link below and sign in with your email@utah.edu, once in you will be redirected via single sign on and can sign in with your unid and password.

A secondary book is Python for Data Analysis. This book is a thorough resource to learn many commands needed for exploratory data analysis, cleaning and transforming data with Python, Pandas, and Numpy. This course will use many commands and techniques covered in the book.

$$DSB$$

Foster Provost and Tom Fawcett, Data Science for Business, O'Reilly, 2013

https://www.oreilly.com/library/view/data-science-for/9781449374273/

$$PFDA$$

Wes Mckinney, Python for Data Analysis, O'Reilly, 2017

https://learning.oreilly.com/library/view/python-for-data/9781491957653/

$$HAML$$

Aurélien Géron, Hands on Machine Learning with Scikit-learn, Keras and Tensorflow, O'Reilly, 2021

https://learning.oreilly.com/library/view/hands-on-machine-learning/9781492032632/

Technology Requirement

Students are required to have access to a laptop or desktop. The laptop or desktop must be equipped with a reasonable up-to-date operating system (OS) version, internet-enabled, up-to-date anti-virus software, sufficient RAM and hard-drive space.

The School of Business and the IS Program strongly recommend purchasing a Microsoft Windows Laptop or Desktop. This is due to ease of working with business applications and software needed for classes in Accounting, Finance, Information Systems, and Business Analytics.

LLMs such as chatGPT

You may use LLMs such as chatGPT to complete coding for assignments, you may NOT use it to generate any writing such as observations about the data or modeling results. All writing must be your own.

You may NOT use it on quizzes or exams.

CANVAS

Course presentations, syllabus, assignments, recommended readings, grades, etc. are available on CANVAS. Please see the course schedule online and on the syllabus for details regarding when and where to submit your assignments. In order to access Canvas, you need to have an active University Network ID (UNID). For more information, go to https://go.utah.edu/cas/login.

Assignments

All assignments are due at 11.59 pm on the due date as posted in Canvas (unless otherwise specified). If you know ahead of time that you will have difficulty meeting an assignment deadline, notify the instructor as soon as possible. Exceptions may be made on a case-by-case basis for reasons such as sickness and family emergencies. For late assignments, the final assignment score will be dropped by 20% credit each day (up to 3 days late) unless prior approval for an extension has been obtained.

Submitting an assignment late (even 1 minute late) will result in the late penalty being applied. No exceptions will be made for this rule. Submit early to ensure you don't receive a deduction.

When using computers and technology, technical difficulties often arise. This is especially pertinent for online classes. Please accommodate for this likely possibility and plan to complete your work before the due date! Any excuses for late submissions due to such reasons are not acceptable.

The syllabus and schedule are subject to change. Any communications will be published as early as possible; however, you are responsible to check CANVAS on a regular basis to ensure your assignments get turned in on time.

Quizzes

Quizzes will be timed and held in class on Tuesday mornings using LockDown Browser. Students must bring a compatible laptop with LockDown Browser installed and working. Quizzes are open book and open notes, but ChatGPT and other LLMs are not permitted. The late penalty does not apply to quizzes, and missed quizzes will receive a 0 unless otherwise approved by the instructor.

Exams

Exams will be held in person on Wednesday of the scheduled exam week and proctored using LockDown Browser. Exams are closed book, closed internet, and closed LLMs (such as ChatGPT). Students may use one single-sided sheet of handwritten notes; no other notes or reference materials are permitted. Students must bring a compatible laptop with LockDown Browser installed and working. Missed exams will result in a 0.

Exam 2 is not comprehensive. There is no comprehensive final exam during the final exam week (Dec14-18).

Class Contribution

Contribution includes preparation, thoughtful participation in case analysis and demonstrations, constructive peer feedback, and professional engagement during applied work. Attendance alone does not constitute contribution, but students cannot contribute consistently when absent.

Students will use a Google Form to document their contributions during class. Entries should briefly and accurately describe a contribution the student actually made and must be submitted according to the instructions provided in class. Form submissions create a record for assessment but do not automatically earn credit; the instructor may evaluate their relevance, quality, and accuracy.

Submitting the form without making the reported contribution, or otherwise falsifying or materially misrepresenting a contribution, is an academic integrity violation. Such conduct may result in a score of zero for the entire class-contribution category for the semester and may be referred for action under University Policy 6-410.

At the end of the semester, each student's documented contribution scores will be averaged. A curve based on the class's contribution results will then be applied to determine final contribution scores. This approach recognizes meaningful participation while accounting for differences in the number and type of contribution opportunities available during the semester.

When an absence prevents participation, a student may submit the course's Missed Attendance Form to request that the absence be excused. If approved, the student will receive the class's average contribution points for that class meeting. This option is intended for occasional, legitimate absences and should be used sparingly. A student may use the Missed Attendance Form no more than three times during the semester. Submitting the form does not automatically excuse an absence, and students remain responsible for missed course material and assignments.

Academic Dishonesty (Cheating)

All assignments are to be completed on an individual basis unless informed otherwise. Do not assume that it is okay to work with someone else on an assignment, unless explicit permission was given. Sharing answers on an assignment without permission could result in an academic integrity violation and possible disciplinary consequences. Some assignments will require application of the concepts covered in class/reading assignments by utilizing relevant software, while others may just require a written response. Please contact the instructor in advance of a due date if you have any questions.

The department has a zero-tolerance policy for cheating behavior. If you are caught cheating on one of my assignments or exams in this class, a formal academic penalty will be imposed. Possible sanctions for cheating include redoing an assignment, a grade reduction, a failing grade for the assignment, or a failing grade for the course. The severity of the penalty will depend on the circumstances and is at the sole discretion of your instructor and the Department of Operations and Information Systems. Cheating includes, but is not limited to, using another student’s work as well as supplying another student with work that is not their own. All parties involved in academic dishonesty will be penalized for cheating.

University of Utah Student Code

The Student Code is spelled out in the Student Handbook. Students have specific rights in the classroom as detailed in Article III of the code. The code also specifies proscribed conduct (Article XI) that involves cheating on tests, plagiarism, and/or collusion, as well as fraud, theft, etc. Students should read the Code carefully to become aware of these issues. Students will receive sanctions for violating one or more of these proscriptions. The faculty will enforce the code. Students have the right to appeal such action to the Student Behavior Committee. The Student Code is provided in detail on the following University of Utah web page: www.admin.utah.edu/ppmanual/8/8-10.html

Drop/Withdrawal Policy

Once a student is officially enrolled and committed to attend class, the student must officially drop the class by the deadline. For Fall 2026 semester-length classes, the last day to drop (delete) the class is Friday, September 4, and the last day to withdraw is Friday, October 30. See the University of Utah Fall 2026 Academic Calendar for official dates and updates. If the class is not officially dropped, the student will be charged full tuition and may receive a failing grade. Although some departments dismiss students from classes for non-attendance, students are responsible for officially dropping any classes for which they are registered but not attending.

Beginning the eleventh calendar day and continuing through the midpoint students may withdraw from a class or the university without instructor/department permission. A "W" grade is recorded on the transcript and appropriate tuition and fees are assessed. After the midpoint of the term, students may petition the deadline for withdrawal if they have a nonacademic emergency. The Petition for Consideration of Exception to the Withdrawal Policy form may be obtained from the appropriate dean's office. Submit the petition and supporting documentation to the dean's office by the last day of class for the course.

Incompletes

University policy states: “An Incomplete grade can be given for work not completed due to circumstances beyond your control. You must be passing the course and have completed at least 80% of the required coursework. Arrangements must be made between you and the instructor concerning the completion of the work. You may not retake a course without paying tuition. If you attend class during a subsequent term, in an effort to complete the coursework, you must register for the course. Once the work has been completed, the instructor submits the grade to the Registrar's Office. The "I" will change to an "E" if a new grade is not reported within one year. A written agreement between you and the instructor may specify the grade to be given if the work is not completed within one year. Copies of the agreement are kept by the instructor and the academic department. If you graduate before a new grade is reported, the "I" remains on your record and will not count towards graduation or the calculation of your grade point average.”

Students with Disabilities

The University of Utah, David Eccles School of Business seeks to provide equal access to its programs, services and activities for people with disabilities. If you will need accommodations in this class, reasonable prior notice needs to be given to the instructor and to the Center for Disability Services, http://disability.utah.edu/, 160 Olpin Union Building, 581-5020 (V/TDD) to make arrangements for accommodations. All written information in this course can be made available in alternative format with prior notification to the Center for Disability Services. CDS will work with you and the instructor to make arrangements for accommodations.

University Safety Statement

The University of Utah values the safety of all campus community members. To report suspicious activity or to request a courtesy escort, call campus police at 801-585-COPS (801-585-2677). You will receive important emergency alerts and safety messages regarding campus safety via text message. For more information regarding safety and to view available training resources, including helpful videos, visit safeu.utah.edu.

Addressing Sexual Misconduct

Title IX makes it clear that violence and harassment based on sex and gender (which Includes sexual orientation and gender identity/expression) is a civil rights offense subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, color, religion, age, status as a person with a disability, veteran's status or genetic information. If you or someone you know has been harassed or assaulted, you are encouraged to report it to the Title IX Coordinator in the Office of Equal Opportunity and Affirmative Action, 135 Park Building, 801-581-8365, or the Office of the Dean of Students, 270 Union Building, 801-581-7066. For support and confidential consultation, contact the Center for Student Wellness, 426 SSB, 801-581-7776. To report to the police, contact the Department of Public Safety, 801-585-2677(COPS).

Undocumented Student Support Statement

Immigration is a complex phenomenon with broad impact—those who are directly affected by it, as well as those who are indirectly affected by their relationships with family members, friends, and loved ones. If your immigration status presents obstacles to engaging in specific activities or fulfilling specific course criteria, confidential arrangements may be requested from the Dream Center. Arrangements with the Dream Center will not jeopardize your student status, your financial aid, or any other part of your residence. The Dream Center offers a wide range of resources to support undocumented students (with and without DACA) as well as students from mixed-status families. To learn more, please contact the Dream Center at 801.213.3697 or visit dream.utah.edu.

Student Mental Health Resources

- Rates of burnout, anxiety, depression, isolation, and loneliness have noticeably increased during the pandemic. If you need help, reach out for campus mental health resources, including counseling, trainings and other support.
- Consider participating in a Mental Health First Aid or other wellness-themed training provided by our Center for Student Wellness and sharing these opportunities with your peers, teaching assistants and department colleagues.

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*Grading

	Percentage
10 Topic Survey Feedback	2
12 Quizzes	20
11 Homeworks	20
Exam 1	24
Exam 2	24
1 Individual Project	10
Total	100

*Notes:

1. The number of assignments for each category may change.
2. Quizzes are timed, held in class, and administered using LockDown Browser.
3. Exams are held in person on Wednesday of the scheduled exam week and proctored using LockDown Browser. They are closed book, closed internet, and closed LLMs. One single-sided sheet of handwritten notes is permitted.
4. Extra Credit Available for
    - Course Evaluation
    - Competitions
    - DataCamp Course Completion

Grading Scale
Final Letter Grades are determined after reviewing course grade distributions to ensure grades are within the expected average grading policy range.

David Eccles School of Business -- Statement of Grading Policy
Grading provides feedback to students on how well they have mastered the content and learning objectives of a particular course to allow students to capitalize on strengths and work to improve weaknesses through future courses of action. The DESB grading policy is intended to ensure grades offer reliable feedback regarding student performance, and to ensure fairness and consistency across the School. The faculty member is responsible for arriving at a grade for each student that the faculty member believes appropriately reflects the student's mastery of the course material and learning objectives. The faculty member will then consider the class' overall performance in terms of School guidelines. These guidelines are provided to ensure that grading, on average for the School as a whole, is sustained at a reasonable level over time. The guidelines are as follows:

COURSE LEVEL GUIDELINE
1000-2000 2.8-3.2
3000-3990 3.0-3.4
4000-5990 3.1-3.5
6000-6990

If students have a concern about their grade in a particular course, they should consider whether it reflects an accurate evaluation of their mastery of the course material and learning objectives, in terms of the above descriptors. If they need clarification of the instructor's evaluation, they should meet with the instructor to obtain additional information and feedback. If after doing so, they believe their grade was arrived at in an inappropriate manner, they may pursue an appeal through the School's appeals process as described in Section 5.15 of the University of Utah Code of Student Rights and Responsibilities (Policy 6-400).

Tentative Course Schedule (subject to revision)

MS = Weekly Module Survey

Module	Topic	Readings	Deliverables
1 (8/24-8/30)	Week 1 - Welcome, introductions, and getting started	Readings (Welcome to Colaboratory); DSB Ch. 1	HW 1; Q1; MS 1
2 (8/31-9/6)	Week 2 - Data mining introduction and exploratory data analysis	DSB Ch. 2; $PFDA$ Ch. 5, 6.1-6.2, 9	HW 2; Q2; MS 2
3 (9/7-9/13)	Week 3 - Introduction to classification	DSB Ch. 2; $PFDA$ Ch. 9	HW 3; Q3; MS 3
4 (9/14-9/20)	Week 4 - Naive Bayes	DSB Ch. 3, 7; $PFDA$ Ch. 10	HW 4; Q4; MS 4
5 (9/21-9/27)	Week 5 - Numeric prediction and descriptives	DSB Ch. 5, 7, 9	HW 5; Q5; MS 5
6 (9/28-10/4)	Week 6 - Linear regression and regression trees	DSB Ch. 4	Q6; MS 6
7 (10/5-10/9)	Week 7 - **Exam 1: Wednesday, October 6 (in person with LockDown Browser)**	DSB Ch. 4	HW 6
10/10-10/18	**Fall Break - no class Tuesday, October 13**		
8 (10/19-10/25)	Week 8 - Support vector machines and grid search	DSB Ch. 4	Q8; MS 7

Module	Topic	Readings	Deliverables
9 (10/26-11/1)	Week 9 - Neural networks	DSB Ch. 4	HW 7; Q9; MS 97
10 (11/2-11/8)	Week 10 - K-nearest neighbors and project EDA submission	DSB Ch. 6	HW 8; Q10; MS 9
11 (11/9-11/15)	Week 11 - Clustering	DSB Ch. 6	HW 9; Q11; MS 10
12 (11/16-11/22)	Week 12 - Recommender Engines	DSB Ch. 12	Q12; MS 11
13 (11/23-11/29)	Week 13 - Ensemble methods	DSB Ch. 12	HW 10; MS 12
14 (11/30-12/6)	Week 14 - **Exam 2: Wednesday, December 1 (in person with LockDown Browser)**; project work and discussions		
15 (12/7-12/10)	Week 15 - Final class, project submission, and next steps for learning more about data mining		Final Project Files Submission; HW 11; MS 13